

MODUL 14

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No 1 selection sort

```
package main

import "fmt"

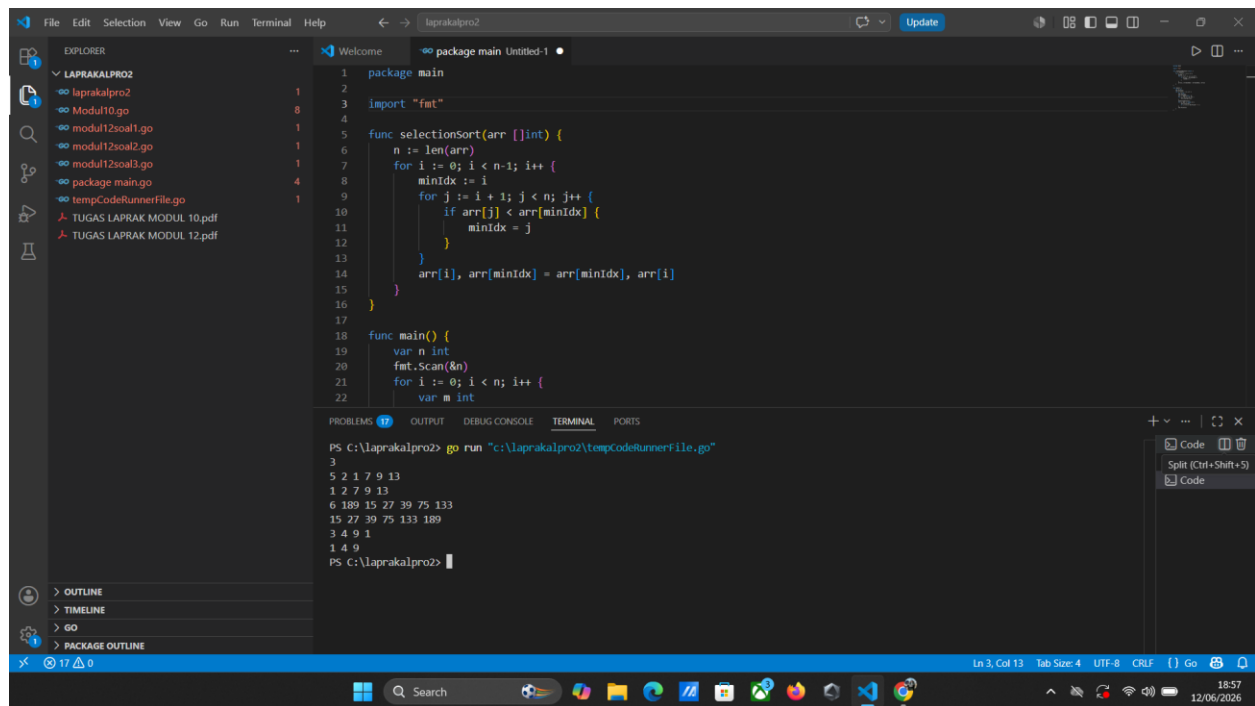
func selectionSort(arr []int) {
    n := len(arr)
    for i := 0; i < n-1; i++ {
        minIdx := i
        for j := i + 1; j < n; j++ {
            if arr[j] < arr[minIdx] {
                minIdx = j
            }
        }
        arr[i], arr[minIdx] = arr[minIdx], arr[i]
    }
}

func main() {
    var n int
```

```
fmt.Scan(&n)

for i := 0; i < n; i++ {
    var m int
    fmt.Scan(&m)
    arr := make([]int, m)
    for j := 0; j < m; j++ {
        fmt.Scan(&arr[j])
    }
    selectionSort(arr)
    for j, val := range arr {
        fmt.Print(val)
        if j < m-1 { fmt.Print(" ") }
    }
    fmt.Println()
}
}
```

Screenshot output



No 2

```
package main
```

```
import (
```

```
    "fmt"
```

```
    "sort"
```

```
)
```

```
func main() {
```

```
    var n int
```

```
    fmt.Scan(&n)
```

```
    for i := 0; i < n; i++ {
```

```
        var m int
```

```
fmt.Scan(&m)

var odds, evens []int

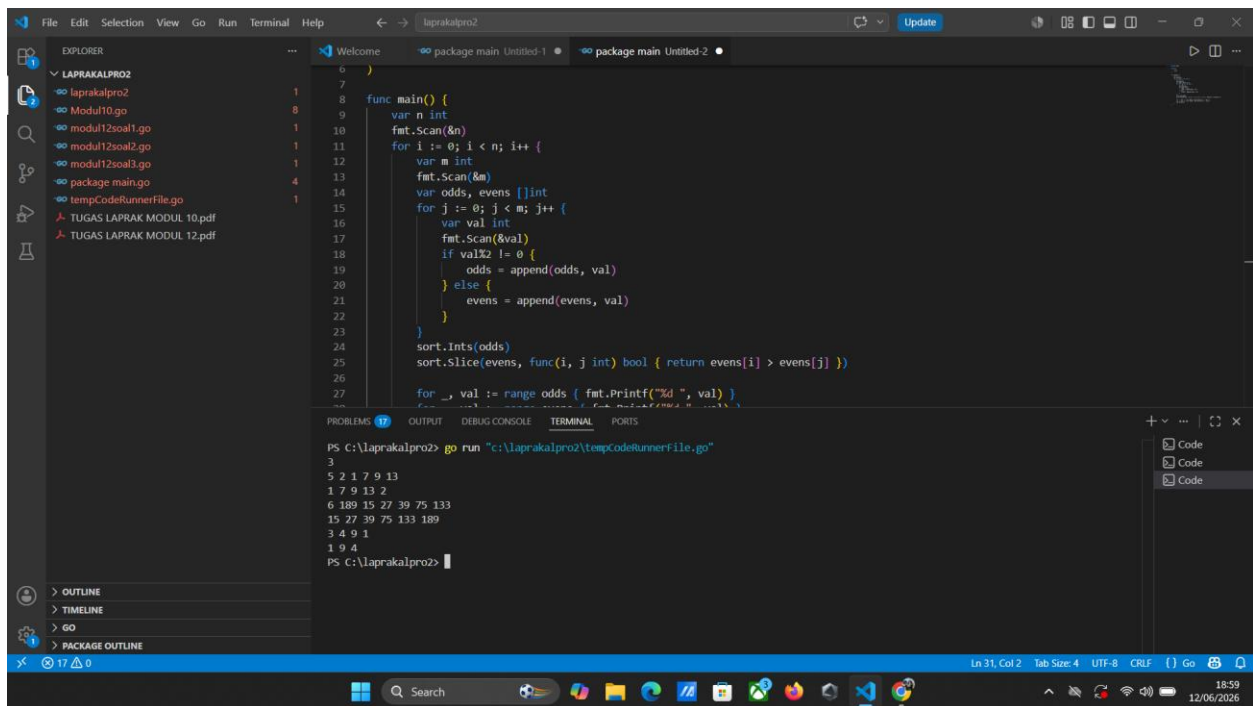
for j := 0; j < m; j++ {
    var val int
    fmt.Scan(&val)
    if val%2 != 0 {
        odds = append(odds, val)
    } else {
        evens = append(evens, val)
    }
}

sort.Ints(odds)

sort.Slice(evens, func(i, j int) bool { return evens[i] > evens[j] })


for _, val := range odds { fmt.Printf("%d ", val) }
for _, val := range evens { fmt.Printf("%d ", val) }
fmt.Println()
}
}
```

Screenshot ouput



No 3

```
package main
```

```
import (
```

```
    "fmt"
```

```
    "sort"
```

```
)
```

```
func main() {
```

```
    var data []int
```

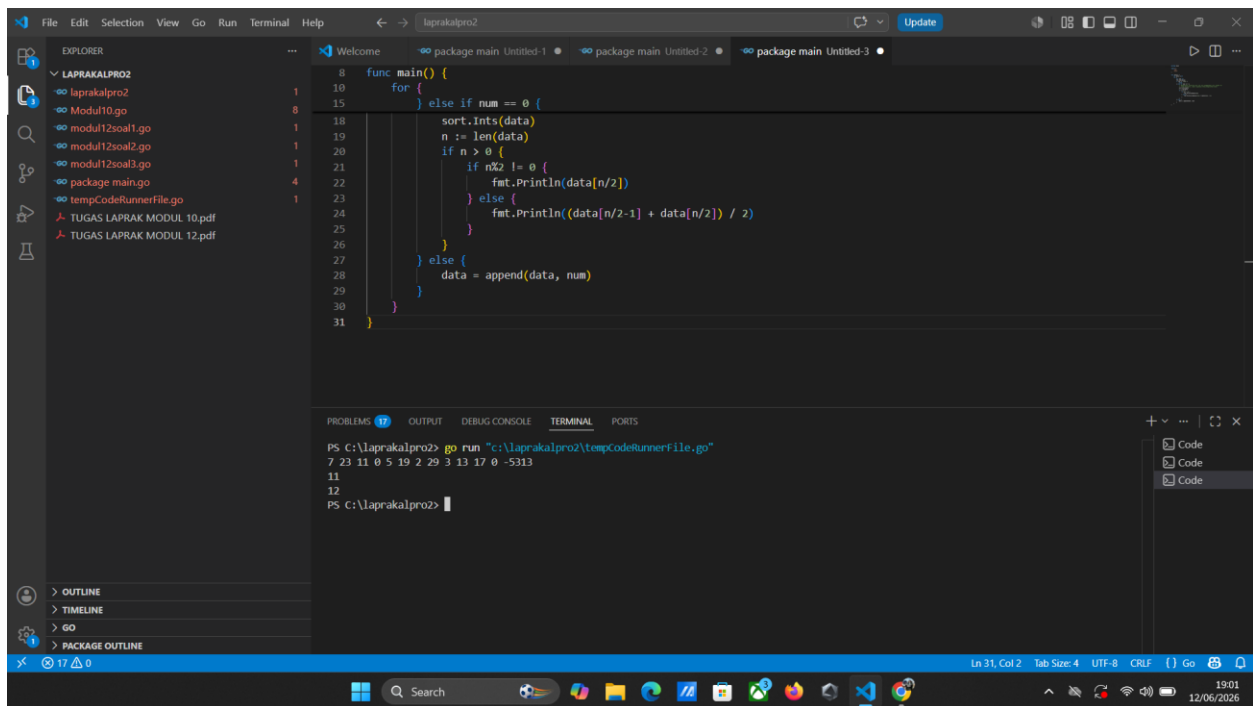
```
    for {
```

```
        var num int
```

```
        fmt.Scan(&num)
```

```
if num == -5313 {  
    break  
} else if num == 0 {  
    sort.Ints(data)  
    n := len(data)  
    if n > 0 {  
        if n%2 != 0 {  
            fmt.Println(data[n/2])  
        } else {  
            fmt.Println((data[n/2-1] + data[n/2]) / 2)  
        }  
    }  
} else {  
    data = append(data, num)  
}  
}  
}
```

Screenshot ouput



No 4

package main

import (

"fmt"

)

func insertionSort(arr []int) {

for i := 1; i < len(arr); i++ {

key := arr[i]

j := i - 1

for j >= 0 && arr[j] > key {

```

        arr[j+1] = arr[j]

        j--
    }
    arr[j+1] = key
}
}

func main() {
    var input []int
    for {
        var n int
        fmt.Scan(&n)
        if n < 0 { break }
        input = append(input, n)
    }

    insertionSort(input)
    fmt.Println(input)

    if len(input) > 1 {
        diff := input[1] - input[0]
        isConstant := true
        for i := 2; i < len(input); i++ {
            if input[i]-input[i-1] != diff {
                isConstant = false
            }
        }
    }
}

```



```

        break
    }
}

if isConstant {
    fmt.Printf("Data berjarak %d\n", diff)
} else {
    fmt.Println("Data berjarak tidak tetap")
}
}
}

```

Screenshot ouput

The screenshot shows a Go IDE with a dark theme. The Explorer panel on the left shows a project named 'LAPRAKALPRO2' with files like 'laprakalpro2', 'Modul10.go', 'modul12soal1.go', 'modul12soal2.go', 'modul12soal3.go', 'package main.go', and 'tempCodeRunnerFile.go'. The main editor displays a Go program with an insertion sort function and a main function. The terminal at the bottom shows the execution of the program, which reads an array of numbers and prints the spacing between them.

```

7 func insertionSort(arr []int) {
8     for i := 1; i < len(arr); i++ {
11         for j := 0; j < i; j++ {
12             if arr[j] > arr[j+1] {
13                 arr[j], arr[j+1] = arr[j+1], arr[j]
14             }
15         }
16         arr[j+1] = key
17     }
18 }
19 func main() {
20     var input []int
21     for {
22         var n int
23         fmt.Scan(&n)
24         if n < 0 { break }
25         input = append(input, n)
26     }
27 }
28 insertionSort(input)
29 fmt.Println(input)
30 }

```

```

PS C:\laprakalpro2> go run "c:\laprakalpro2\tempCodeRunnerFile.go"
31 13 25 43 1 7 19 37 -5
[1 7 13 19 25 31 37 43]
Data berjarak 6
PS C:\laprakalpro2> go run "c:\laprakalpro2\tempCodeRunnerFile.go"
4 40 14 8 26 1 38 2 32 -31
[1 2 4 8 14 26 32 38 40]
Data berjarak tidak tetap
PS C:\laprakalpro2>

```